

Climate

i. Monsoon

*The word '**Monsoon**' is derived from the Arabic word '**Mausim**' which means season. *Traditionally, the 'Monsoon' is related to the change in the pattern of monsoonal winds through which rain occurs in South and Southeastern Asian countries. India receives near-about **75%** of its total precipitation from the southwest monsoon. The monsoon is active from **June to September**, which is very important for agriculture in India. *The monsoon arrives at the southern tip (Kerala) of the Indian peninsula by the first week of June. India's climate has characteristics of both tropical and subtropical climates.



Origin of Indian Monsoon-

Monsoon is a result of **differential rates of heating and cooling of land and sea**. The sun is vertical over the Tropic of Cancer in summer season, and the northern regions of the Indian landmass get heated. This leads to the formation of low pressure conditions over the Indian subcontinent. On the other hand, the Indian Ocean is colder, so high pressure is generated, leading to northward shifting of Inter Tropical Convergence Zone. ***The Inter Tropical Convergence Zone (ITCZ)** is a low-pressure belt near the equator where trade winds of the northern and the southern hemispheres converge. *Due to the northward shift of the ITCZ, the trade winds of the southern hemisphere cross the equator between **40°** and

60° E longitudes and starts blowing from the southwest to the northeast under the influence of the **Coriolis** force. It becomes southwest monsoon. *In winter, the ITCZ moves southward, and so the reversal of winds from northeast to south and southwest takes place. They are called **northeast monsoons**. *The sudden onset of the moisture-laden winds associated with violent thunder and lightning is often termed as the '**burst**' of the monsoon. The monsoon may burst in the first week of June in the coastal area of Kerala.

*The monsoon approaches the Indian landmass in two main branches:

- (i) The Arabian Sea branch
- (ii) The Bay of Bengal branch

*The Tibetan Plateau has a very broad impact on the Indian climatic pattern.

*Rainfall received from the southwest monsoons is seasonal in character, occurring between June and September. Monsoonal rainfall is largely governed by relief or topography. The monsoon rainfall has a declining trend with increasing distance from the sea. The rainless intervals are known as '**breaks**' in the monsoon.

The Retreat of the Monsoon - The months of October and November are known for **retreating monsoons**.

*The weather in the retreating monsoon is dry in north India whereas it is associated with rain in the eastern part of the Peninsula. The widespread rain in this season is associated with the passage of cyclonic depressions that originate over the Andaman Sea and manage to cross the eastern coast of the Southern Peninsula. These tropical cyclones are very destructive. A bulk of the rainfall on the Coromandel coast is derived from these depressions and cyclones. Such cyclonic storms are less frequent in the Arabian Sea.

Sr. No.	Local Storms of Hot weather Season	Region
(i)	Mango Shower	Kerala, and Coastal areas of Karnataka
(ii)	Blossom Shower	Kerala
(iii)	Norwesters (Kalbaisakhi)	West Bengal and Assam
(iv)	Loo	Northern Plains

***El-Nino** is a complex weather system bringing drought, floods and other weather extremes in different parts of the world. It occurs with the sudden appearance of warm currents off the coast of Peru (warm equatorial current replaces cold Peruvian current).